Voice-over Script · Review – I

NASH Dhabha & Bakes: Masala Mitra (Phase 1 proposal)

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Course	25ARE471 Introduction to Generative AI
Guided by	Prof. Rakesh Kumar
Slides	19
Estimated time	8:10 min at ~145 words per minute

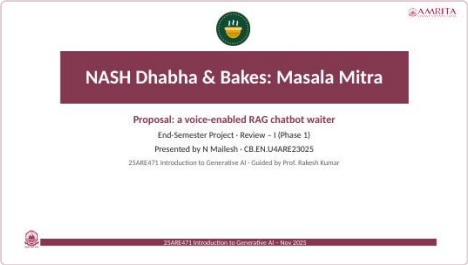
Delivery tips

Speak slightly slower than normal and pause briefly at each slide change. The same text is in each slide's speaker notes (Presenter View). On the prototype slide, you can switch to the live app for a short demo.

SLIDE 1

0:00 – 0:27 · ~27 s

Title

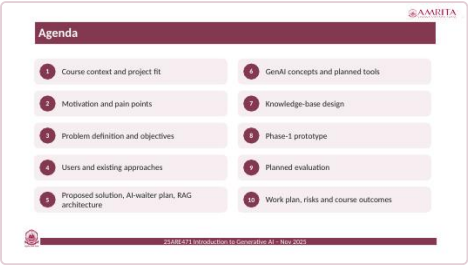


Good morning. I'm N Mailesh. For my Introduction to Generative AI project, I'm proposing Masala Mitra: a voice-enabled, retrieval-augmented chatbot that works like a waiter for a restaurant called NASH Dhabha and Bakes. In this first review I'll cover the problem, my objectives, the planned design, how it maps to the course, and the prototype I've built so far.

SLIDE 2

0:27 – 0:54 · ~27 s

Agenda

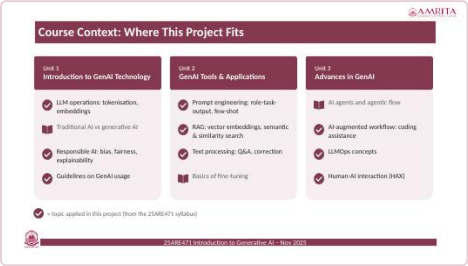


Here's the agenda. I'll start with how the project fits the course, then the motivation, problem and objectives. Next come the users and existing approaches, my proposed solution and its RAG architecture, the GenAI concepts and tools I plan to use, the knowledge-base design, and the phase-one prototype. I'll finish with the evaluation plan, work plan, risks and course-outcome mapping.

SLIDE 3

0:54 – 1:19 · ~25 s

Course Context: Where This Project Fits

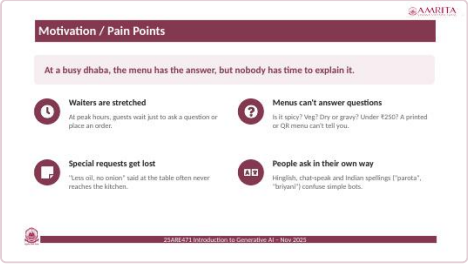


The course has three units. This project applies most of them. From Unit 1: tokenisation and embeddings, and responsible AI. From Unit 2: prompt engineering, retrieval-augmented generation with vector similarity search, and question answering. From Unit 3: the AI-augmented coding workflow, LLM-ops style evaluation, and human-AI interaction. The ticks mark the topics I apply.

SLIDE 4

1:19 – 1:44 · ~25 s

Motivation / Pain Points

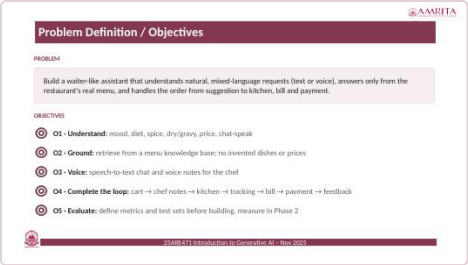


The motivation comes from everyday dhaba experience. Waiters are stretched at peak hours. A menu card can't answer simple questions like 'is it spicy' or 'is it dry or gravy'. Special requests often never reach the kitchen. And customers ask in their own way, with Hinglish, chat-speak and Indian spellings, which confuses simple chatbots.

SLIDE 5

1:44 – 2:10 · ~26 s

Problem Definition / Objectives

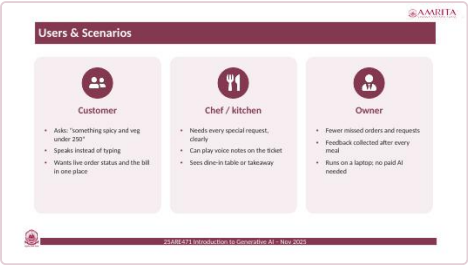


The problem: build a waiter-like assistant that understands natural, mixed-language requests by text or voice, answers only from the real menu, and handles the whole order. My five objectives: understand requests, ground every answer in the menu, support voice, complete the full ordering loop, and define the evaluation up front so I can measure it in phase two.

SLIDE 6

2:10 – 2:33 · ~23 s

Users and Scenarios



There are three users. The customer wants to ask naturally, perhaps by voice, and follow the order in one place. The chef needs every special request clearly, including voice notes, on the ticket. The owner wants fewer missed orders, feedback after every meal, and something that runs without expensive AI services.

SLIDE 7

2:33 – 2:58 · ~25 s

Existing Approaches and the Gap



Here's the gap. Printed and QR menus are accurate but can't answer questions. A generic chatbot can answer, but it isn't grounded in the restaurant's real menu, so it can invent dishes or prices. None of them combine voice, Hinglish understanding, chef notes and the full order flow. That combination is what Masala Mitra proposes.

SLIDE 8

2:58 – 3:25 · ~27 s

Proposed Solution: Masala Mitra

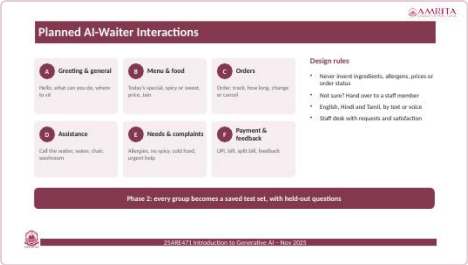


The proposed solution is Masala Mitra. It greets customers and asks their mood, then suggests dishes. It answers questions like a waiter, using only the real menu. It supports voice chat, and voice notes for the chef. The website adds a filterable menu and a kitchen display, and the chat itself tracks the order live and brings the bill and payment.

SLIDE 9

3:25 – 4:03 · ~38 s

Planned AI-Waiter Interactions

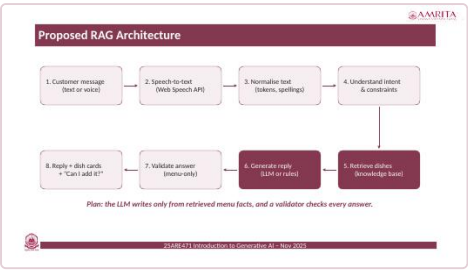


Beyond menu questions, Masala Mitra is planned as a full AI waiter. It covers six groups of interactions: greetings, menu and food questions, orders, assistance like calling the waiter or asking for water, special needs and complaints like allergies, and payment and feedback. Two design rules guide it. It must never invent ingredients, allergens, prices or order status, and when it isn't sure, it hands over to a staff member. It will work in English, Hindi and Tamil, with a staff desk for requests and satisfaction.

SLIDE 10

4:03 – 4:31 · ~28 s

Proposed RAG Architecture

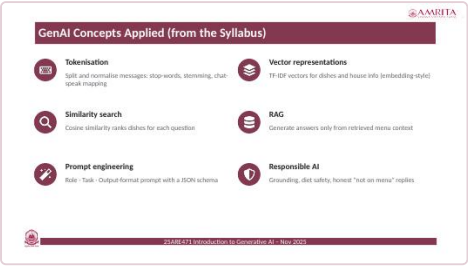


This is the planned architecture. A message comes in as text or voice; voice is converted by speech-to-text. The text is normalised, then the intent and constraints are understood. The retriever finds matching dishes in the knowledge base. A language model, or a rule-based writer, generates the reply only from those dishes. A validator checks the answer before the customer sees it.

SLIDE 11

4:31 – 4:55 · ~24 s

GenAI Concepts Applied

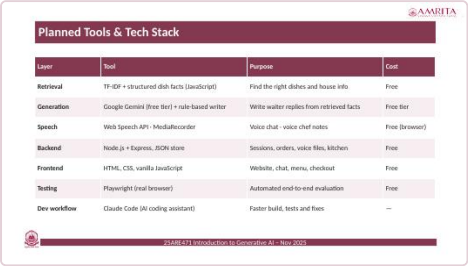


These syllabus concepts are applied directly. Tokenisation splits and normalises each message. Vector representations: dishes and house info become TF-IDF vectors, and cosine similarity ranks them. That's the retrieval in RAG. Prompt engineering follows the Role, Task and Output-format pattern with a JSON schema, and responsible-AI principles guide grounding and diet safety.

SLIDE 12

4:55 – 5:25 · ~30 s

Planned Tools and Tech Stack



The stack is deliberately free. Retrieval uses TF-IDF and structured dish facts in JavaScript. Generation uses Google Gemini's free tier, with a rule-based writer as a fallback. Speech uses the browser's Web Speech API and MediaRecorder. The backend is Node.js and Express, the frontend is plain HTML, CSS and JavaScript, testing uses Playwright in a real browser, and I use Claude Code as an AI coding assistant.

SLIDE 13

5:25 – 5:55 · ~30 s

Knowledge-Base Design

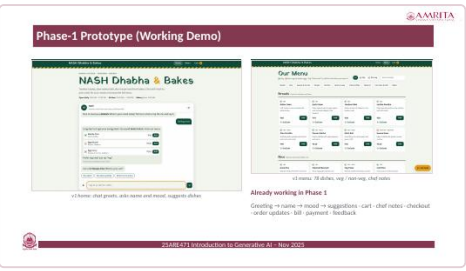


The knowledge base is the restaurant's own menu: ten sections from breads to bakes, plus house-information pages on timings, payment, the veg kitchen, chef notes, Jain food and GST. Each dish records its name, category, price, veg or non-veg, spice level, a description and mood tags. In phase two I plan to add texture, main ingredient and pairings, so it can answer like a real waiter.

SLIDE 14

5:55 – 6:25 · ~30 s

Phase-1 Prototype

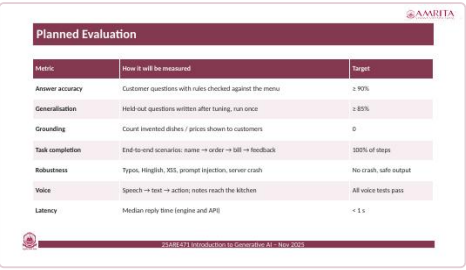


I already have a working phase-one prototype. The chat greets the customer, asks their name and mood, and suggests dishes from the menu. The menu has seventy-eight veg and non-veg dishes with chef notes, and the full flow works: cart, checkout, order updates, bill, payment and feedback. It's a single-file version that runs in the browser; phase two adds the backend, Gemini and much better understanding.

SLIDE 15

6:25 – 6:50 · ~25 s

Planned Evaluation



I'm defining the evaluation now, before building. Answer accuracy will be measured on customer questions, with each answer checked against the menu, targeting at least ninety percent. Generalisation uses held-out questions written after tuning. Grounding means zero invented dishes or prices. Task completion uses end-to-end scenarios, plus robustness checks, voice tests and latency under one second.

SLIDE 16

6:50 – 7:14 · ~24 s

Work Plan

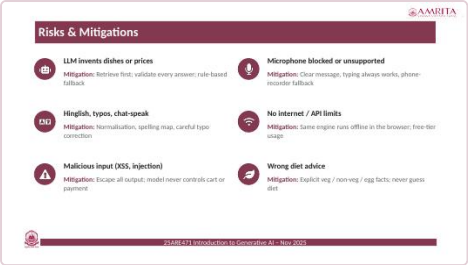


Here's the work plan. Phase one, the prototype, is done. Next I'll add the backend with sessions, orders and a kitchen display; then the waiter brain with dish facts and chat-speak understanding; then voice chat and voice chef notes; then Gemini with guardrails; and finally evaluation, the report and the LinkedIn video.

SLIDE 17

7:14 – 7:40 · ~26 s

Risks and Mitigations

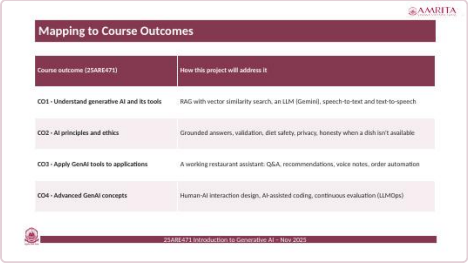


The main risks, and how I'll handle them. The language model might invent dishes, so I retrieve first and validate every answer. The microphone may be blocked, so typing always works and there's a fallback. Hinglish and typos need normalisation. If there's no internet, the engine runs offline. Malicious input is escaped, and diet facts are never guessed.

SLIDE 18

7:40 – 8:01 · ~21 s

Mapping to Course Outcomes



Finally, the mapping to course outcomes. CO1 is covered through RAG, similarity search, an LLM, and speech technology. CO2, ethics, through grounded and validated answers and diet safety. CO3 is the working restaurant assistant itself. CO4 through human-AI interaction design, AI-assisted coding and continuous evaluation.

SLIDE 19

8:01 – 8:10 · ~9 s

Thank You



Thank you. I'd welcome your questions and feedback on the plan before I move into phase two.